RAHUL SHAH

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PROFESSIONAL SUMMARY

Electronics and Telecommunication Engineering graduate focused on Java backend development using Spring Boot, REST APIs, and relational databases. Developed projects involving concurrency control, distributed state management, authentication, and transactional workflows. Seeking an entry-level Java Backend Developer or Software Engineer role.

TECHNICAL SKILLS

Languages	Java, SQL, C++, Python
Backend	Spring Boot, Spring Security, Spring AOP, REST APIs, Spring Data JPA, Hibernate
Databases	MySQL, PostgreSQL, Redis
Messaging	Apache Kafka
Developer Tools	Git, GitHub, Maven, Docker
Core CS	Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design (HLD, LLD)
Security	JWT, OAuth 2.0

PROJECTS

FlashSeat – Flash-Sale Ticket Booking System
Java 21, Spring Boot 3, Redis, PostgreSQL, Lua, SSE, Docker

- Developed a Redis-backed traffic control mechanism using sliding-window counters and a FIFO virtual waiting room to manage request bursts during high-demand ticket sales.
- Implemented queue sequencing using Redis Sorted Sets and delivered real-time queue position updates to clients through Server-Sent Events (SSE).
- Implemented Redis Lua scripts for atomic multi-seat allocation checks, reducing race conditions during concurrent reservation requests.
- Integrated PostgreSQL transactions for confirmed bookings and Redis-based temporary seat locks, enabling seat availability checks across persistent and in-memory state.

FluxLimit – Distributed API Rate Limiter Engine
Java 21, Spring Boot, Redis, Lua, Spring AOP, Docker

- Developed an API rate limiter using the Token Bucket algorithm, enforcing traffic quotas at the controller layer through a custom `@RateLimit` annotation and Spring AOP.
- Implemented atomic token consumption and lazy refill calculations using Redis Lua scripting, ensuring consistent state updates under concurrent requests without background refill threads.
- Handled burst traffic by returning HTTP 429 (Too Many Requests) responses and maintaining centralized token state in Redis to support stateless application instances.

Library Management System
Java, Spring Boot, MySQL, React

- Developed a full-stack library management application with REST APIs for book discovery, loans, reservations, reviews, and wishlists.
- Implemented authentication and authorization using Spring Security, JWT, and Google OAuth 2.0, along with email-based verification.
- Implemented fine calculation and subscription workflows with Razorpay and Stripe payment integrations, alongside user notification functionality.

EDUCATION

Bachelor of Engineering in Electronics and Telecommunication Engineering **2022 – 2026**
Manjara Charitable Trust's Rajiv Gandhi Institute of Technology, Mumbai

University of Mumbai **CGPA: 8.51/10**